GenMedX

AI Healthcare Agent

Team 1:

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System Design

Data Ingestion & Storage

- Patient data ingested from Google Drive → Google Cloud Storage → BigQuery via Cloud Composer. Structured data curated into BigQuery Views for downstream tasks.

Inference Layer (LangGraph Framework)

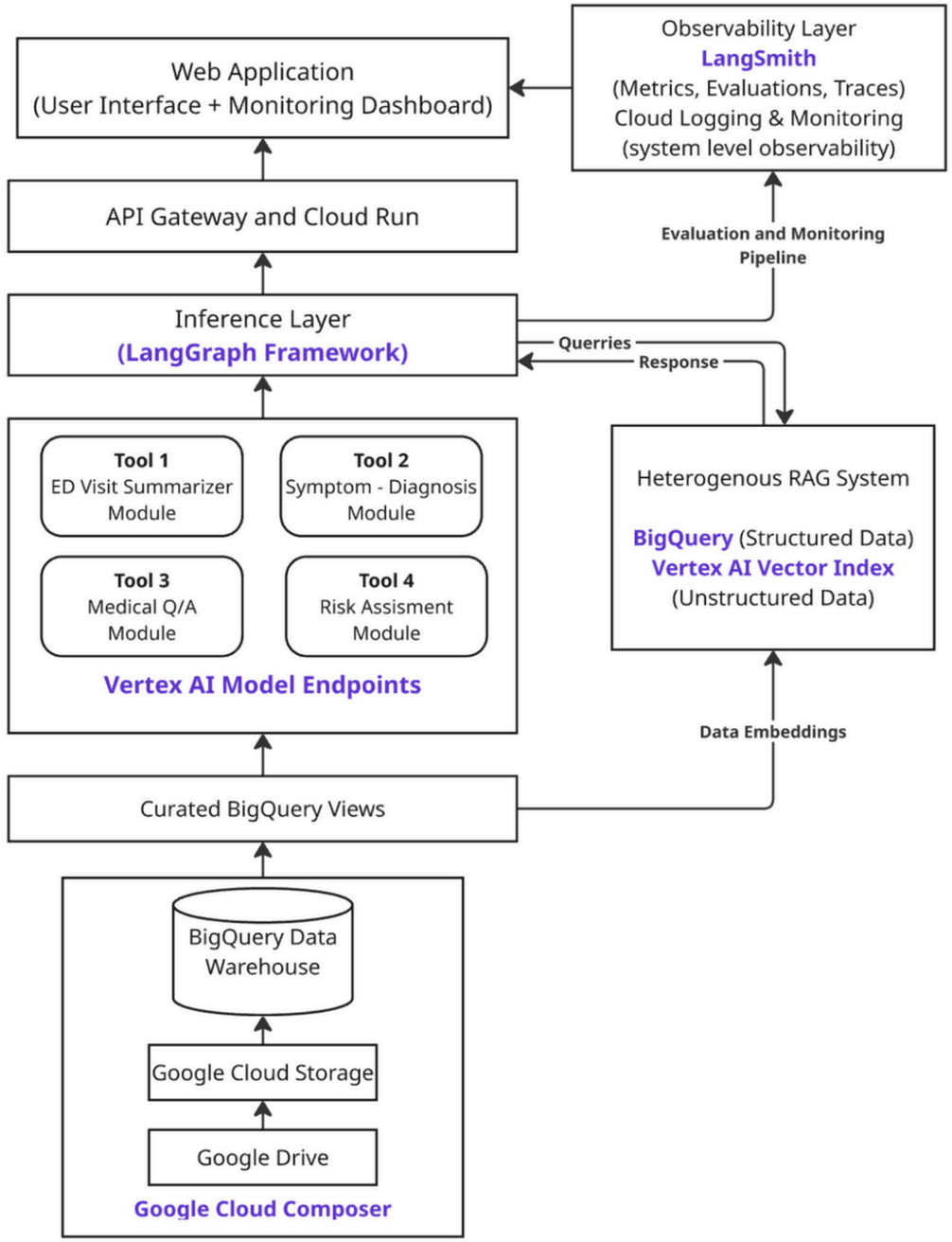
- Orchestrates model calls as functions (tools). Each tool is deployed as a Vertex AI Model Endpoint.

Hetrogenous RAG (Retrieval-Augmented Generation) System

- BigQuery for structured clinical data.
- Vertex AI Vector Index for unstructured notes/documents.

Observability & Monitoring

- LangSmith for trace-level evaluations, metrics, and error analysis.
- Cloud Logging & Monitoring for system health and performance.
- Integrated evaluation pipeline to track model reliability, factuality, and patient-safety metrics.

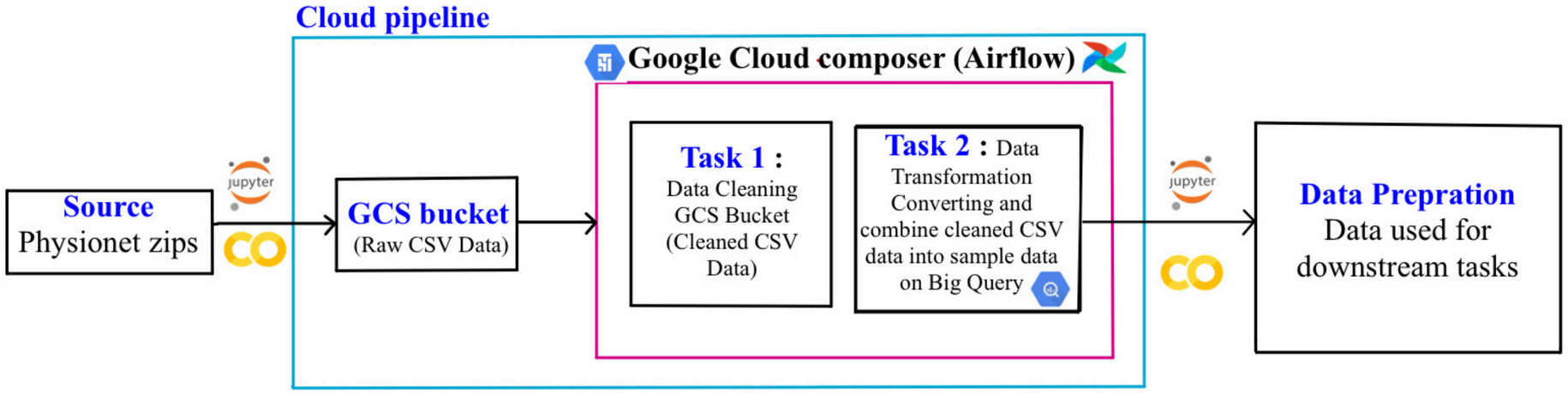


DATA PIPELINE

AUTOMATED CLOUD PIPELINE

Sucessfully processed aproximately 425,000 ER visits and established a structured cloud-based ETL pipeline using Google Cloud Platform

- Google Cloud Storage
- Google Cloud Composer
- Google Big Query



Tool 1: ED Visits Summarizer

DATASET-1

The MIMIC-IV-ED (v2.2) dataset is a large-scale, publicly available resource that contains detailed records of approximately 425,000 emergency department (ED) visits that took place at the Beth Israel Deaconess Medical Center (BIDMC) between 2011 and 2019. It is part of the broader MIMIC-IV data ecosystem and is hosted on PhysioNet, a platform for freely sharing medical data for research.

Table Name	Description	Record Count
edstays	Emergency department visits (unique stays IDs)	~425,087
triage	Triage assessment, vitalsigns, chief complaints	~425,087
vitalsign	Vital Signs recorded throughout the ED stay	~1,564,610
diagnosis	Discharge diagnosis with ICD-10 codes	~899,050
medrecon	Medication reconciliation at admission	~2,987,342
pyxis	Administered medications during ED stays	~1,586,053

Tool 1: ED Visits Summarizer

Models Used

- **LLM (Vertex AI):** Generative model that turns per-stay rows into one short, second-person paragraph.
- **(Logic) LangGraph + pandas:** Orchestration and tabular filtering/grouping

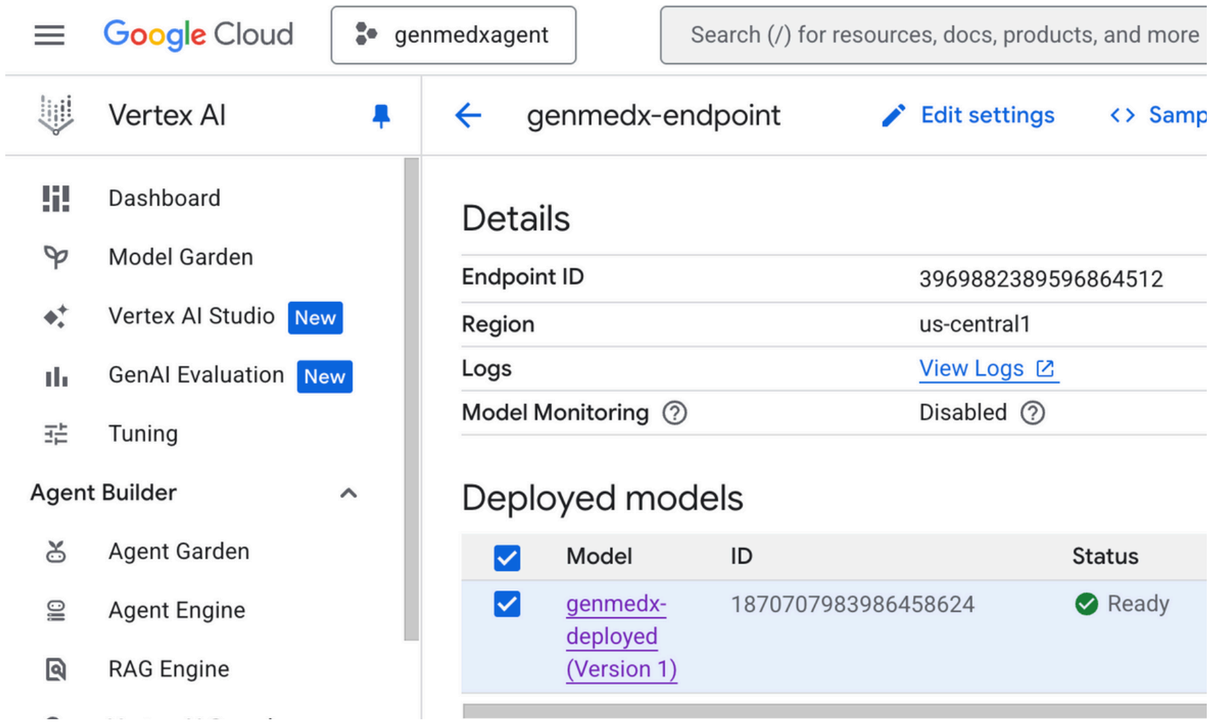
Embedding & Retrieval Workflow

- **Dataset**: MIMIC-ED tables → joined into one CSV (summarizer_data_50_patients.csv, 50 patients).
- **Filter**: Select rows for the input subject_id; group by stay_id to get each visit.
- **Context**: Convert each visit's rows into a compact text block.
- **Generate**: Short prompt → one paragraph per visit (arrival, main issue, diagnosis, outcome).
- **Response**: Join paragraphs → API POST /summarize returns {"predictions":["...", ...]}.
- **Health**: GET /health → 200 OK (readiness check)

Tool 1: ED Visits Summarizer

Input:
Input is patient_id (string/int) plus a free-text question.

Output:
list of short paragraphs, one per stay, joined for convenience




```
(base) spartan@IMS-211MBA genmedx-agent % cat > request.json <<'EOF'
{
  "instances": [
    {"patient_id": "14601430", "question": "Could you summarize my hospital stay history?"}
  ]
}
EOF

(base) spartan@IMS-211MBA genmedx-agent % gcloud ai endpoints predict 3969882389596864512 \
  --region=us-central1 \
  --json-request=request.json
Using endpoint [https://us-central1-prediction-aiplatform.googleapis.com/]
['Stay on 2117-12-15:\nYou arrived at the hospital as a walk-in, complaining of pain in the right lower quad
rant of your abdomen. After an evaluation, you were diagnosed with right lower quadrant abdominal pain and w
ere subsequently stable enough to be discharged home.']
```

Tool 2: Risk Prediction & Tests Recommendation Module

The data used comes from the MIMIC-IV ED database. For this module, we extracted the triage table, which includes patient vitals, chief complaints, and acuity scores. These serve as the foundation for both risk prediction and test recommendation tasks.

Dataset attributes:

- Chief complaint (free-text),
- SBP (mmHg) – systolic blood pressure
- DBP (mmHg) – diastolic blood pressure
- O2 saturation (%)
- Heart rate (bpm)
- Respiratory rate (/min)
- Temperature (°C)
- Pain score (0–10)
- Acuity (ED triage acuity level, 1–5)
- Risk label (derived: 1 = High risk, acuity ⩽ 2; 0 = Low risk, acuity > 2)

Tool 2: Risk Prediction & Tests Recommendation Module

Preprocessing

- Handling missing values: Median imputation for vitals; missingness flags added as explicit features.
- Scaling vitals: StandardScaler applied to normalize values (mean = 0, variance = 1)
- Chief complaint cleaning: Lowercasing, removal of special characters, tokenization.
- Feature engineering: TF–IDF vectorization of complaints (limited to top 500 terms).
- Canonicalization: Normalized test names in recommendations (e.g., “ekg → ECG”, “cxr → Chest X-ray”).
- Integration: Combined structured vitals, flags, and TF–IDF features into a single sparse feature matrix.

Missing values

```
temperature       23415
heartrate         17090
resprate          20353
o2sat             20596
sbp               18291
dbp               19091
pain              12933
acuity             6987
chiefcomplaint       23
dtype: int64
```

Feature matrix

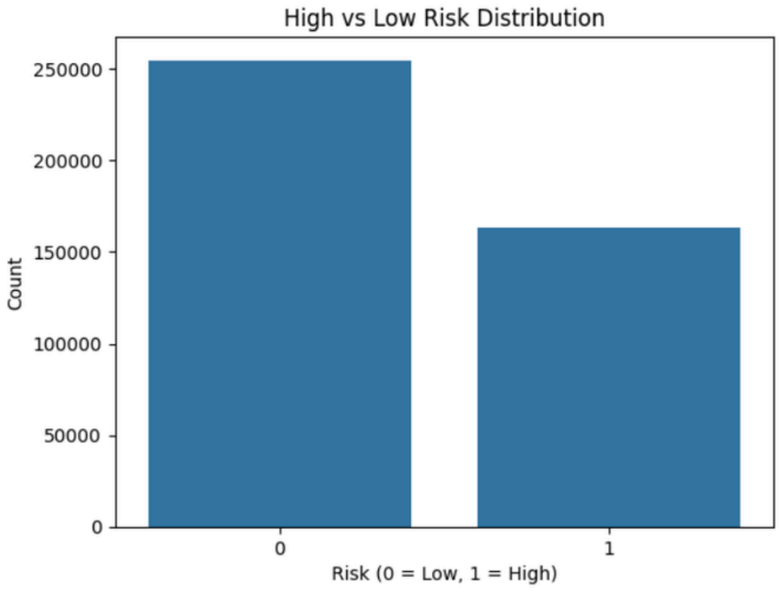
```
Feature matrix shape: (418100, 514)
Target distribution:
 0    254670
1    163430
Name: count, dtype: int64
```

Binary target label

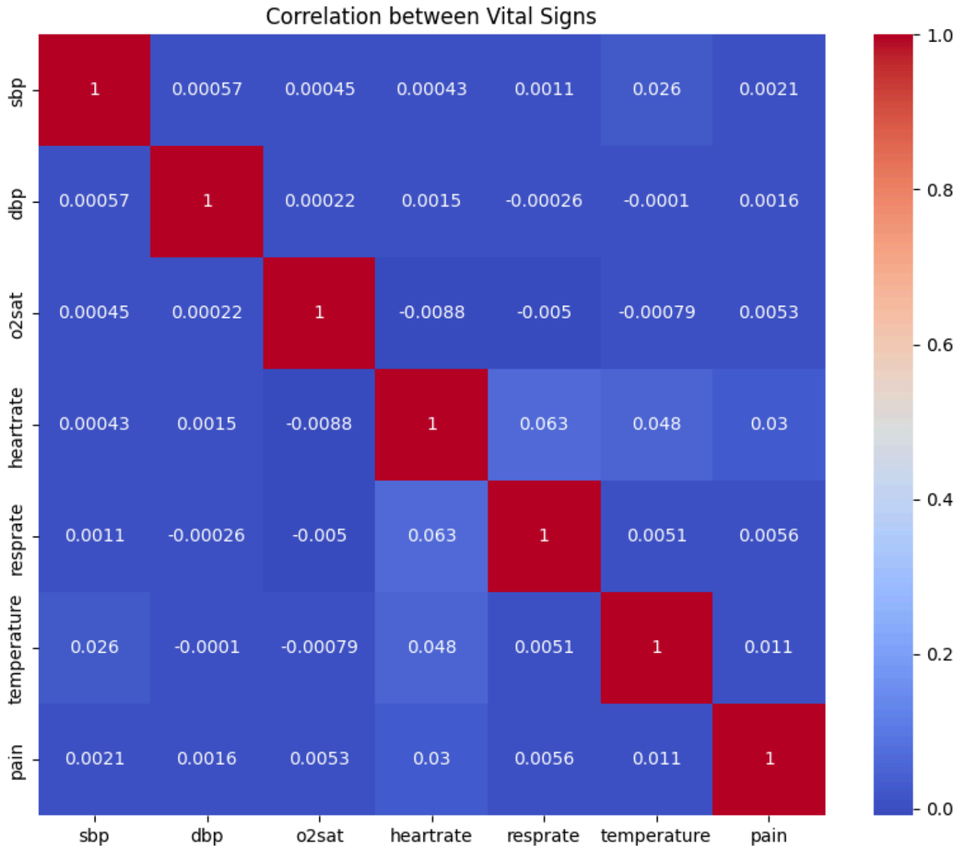
1 = High risk (acuity ≤ 2)

0 = Low risk (acuity > 2)

Tool 2: Risk Prediction & Tests Recommendation Module



Correlation heatmap of vitals



WordCloud for High risk complaints and Low risk complaints

Tool 2: Risk Prediction & Tests Recommendation Module

Models used:

Logistic Regression

- Interpretable baseline model for risk prediction.
- Works on vitals + complaint features.

Random Forest

- Captures nonlinear relationships in patient features.
- Provides feature importance for clinical interpretability.

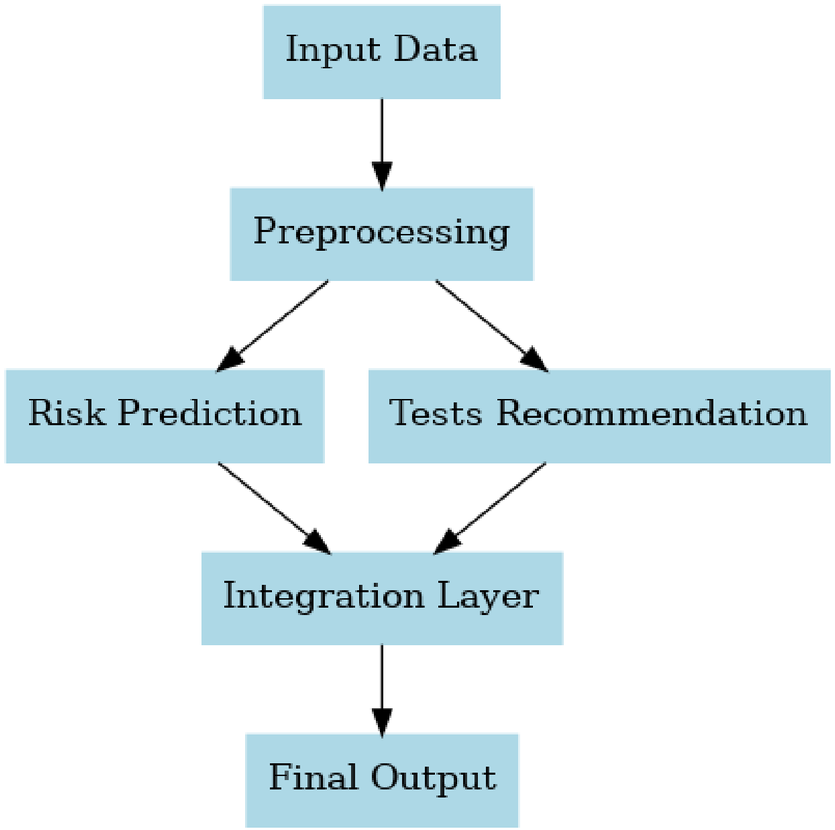
Rule-Based Recommender

- Mapping of complaints and abnormal vitals to standard test sets.
- Guarantees safe and clinically consistent recommendations.
- Serves as a fallback when other models fail.

BioMistral-7B (LLM)

- Biomedical large language model used for test recommendations.
- Generates tests per case, then canonicalized to standard names.

Workflow



Tool 2: Risk Prediction & Tests Recommendation Module

Risk prediction results

	Method	F1 (weighted)	ROC-AUC
0	ML only	0.954629	0.984862
1	Rule-based only	0.589843	N/A
2	ML + Rule-based	0.811798	0.984862
3	BioMistral only (sampled)	0.369484	N/A
4	ML + BioMistral (sampled)	0.374549	N/A (hybrid OR)

Sample Input&Output

```
cc:   "knee pain"
sbp:  88
dbp:  66
hr:   77
rr:   22
o2:   55
temp: 37.1
pain: 6
```

```
{
  "risk": "high",
  "prob": 0.74,
  "rationale": "knee pain + SBP 88 + O2 55",
  "recommended_tests": [
    "ABG/VBG",
    "BMP",
    "Chest X-ray",
    "ECG"
  ],
  "flags": [
    "hypotension",
    "hypoxemia"
  ]
```

Tool 3: Symptom-Diagnosis Module

The dataset, sourced from Hugging Face in CSV/JSON format, contains patient-style symptom notes paired with disease labels. Each record has free-text symptom descriptions (including positive and negated symptoms) with metadata, enabling training and evaluation of models for disease classification.

Dataset attributes:

- **id** – unique identifier for each record
- **disease** – target disease label (e.g., Hypertension, Migraine, Influenza, Pneumonia)
- **note** – free-text symptom description (may include reported symptoms, negated symptoms, onset information, and contextual details such as family history or travel history)
- **created_at** – timestamp indicating when the record was generated or uploaded

Tool 3: Symptom-Diagnosis Module

Models Used

- **SentenceTransformer** (all-MiniLM-L6-v2): Embedding model that converts both knowledge base entries and user queries into 384-dimensional vectors.
- **Gemini 2.5 Pro** (Vertex AI): Generative model that interprets retrieved notes and produces potential condition suggestions with reasoning.

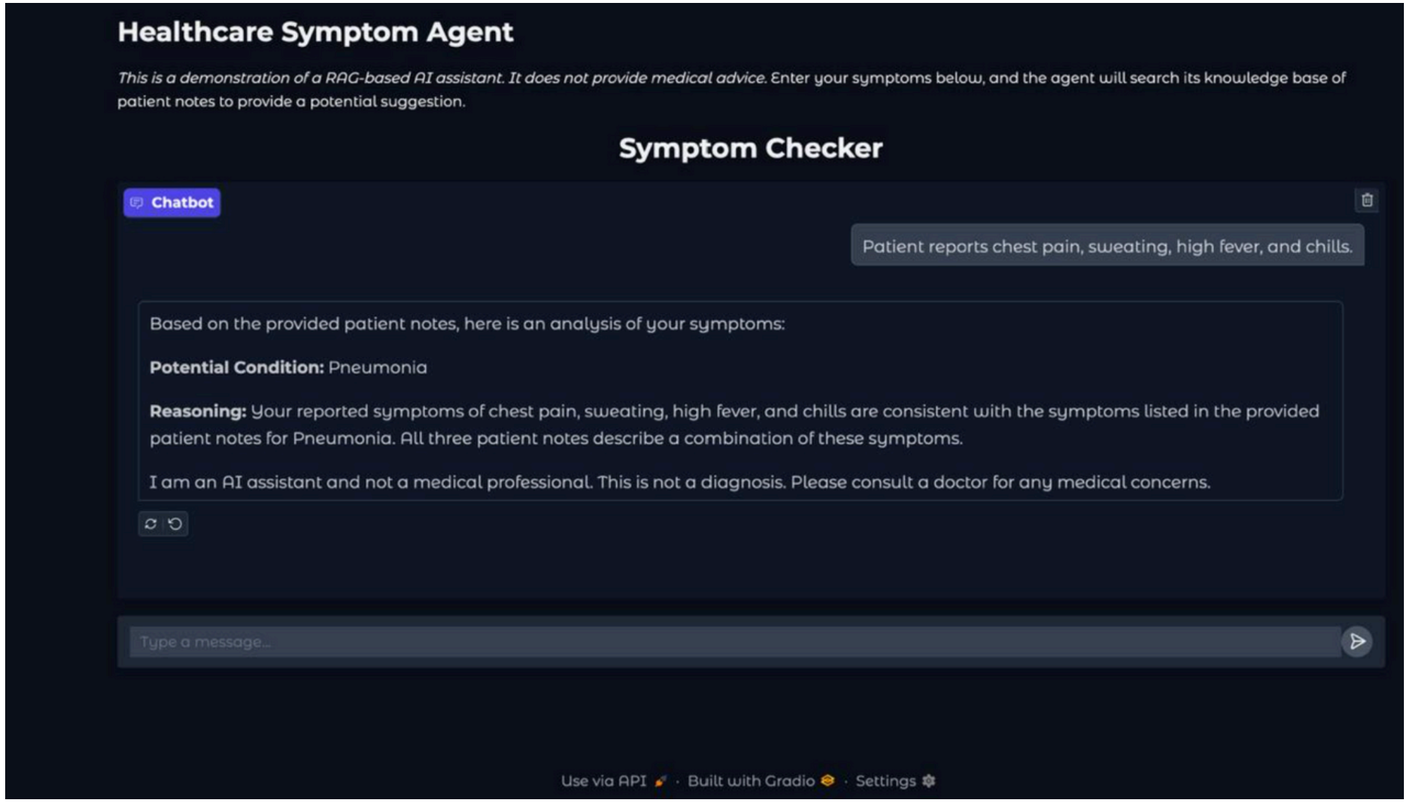
Embedding & Retrieval Workflow

- **Knowledge base:** patient_notes.csv → JSON (knowledge_base.json).
- **Embedding**: Notes converted into 384-dim vectors (all-MiniLM-L6-v2).
- **Vector DB**: Stored in ChromaDB with cosine similarity.
- **Query**: User symptoms embedded for comparison.
- **Retrieval**: Top-3 notes → Gemini 2.5 Pro → condition, reasoning, disclaimer.

Tool 3: Symptom-Diagnosis Module

Input:
Patient's free-text symptom description (e.g., "chest pain, sweating, high fever, and chills").

Output:
Suggested potential condition(s) with reasoning based on retrieved notes, along with a mandatory disclaimer that it is not a medical diagnosis.



Tool 4: MedQA Module

Datasets:

MedQA (USMLE-style): Multiple-choice, exam-like medical knowledge questions spanning core disciplines (e.g., pathophysiology, pharmacology)
Why Used: Strong benchmark for factual correctness and option discrimination; enables Accuracy/Exact-Match and latency comparisons across methods
Preprocessing:

- Unique ID Assignment: Sequential numbering (medqa_mc_001, medqa_mc_002...)
- Direct Format Mapping: Preserved original question-option structure
- Answer Index Standardization: Mapped to A-E format
- Quality Preservation: No filtering needed - all questions valid

PubMedQA: Biomedical Q&A pairs derived from PubMed abstracts with short, evidence-oriented answers.
Why Used: Ideal for testing evidence-grounded generation (RAG): retrieval quality, faithfulness to source text, and concise citation behavior.
Preprocessing:

- Context Integration: Merged research abstract contexts into questions
- Dual Format Generation: Both MC and free-form from same source
- Answer Mapping: "Yes"→"A", "No"→"B", "Maybe"→"C"
- Test Set Filtering: Only processed questions with ground truth labels
- Long Answer Preservation: Maintained detailed explanations for free-form

Method comparision

1. Single LLM

- Multiple Prompting Strategies: Implements 5 different prompting techniques:
  - Zero-shot: Direct question → answer without examples
  - Few-shot: Provides 3 medical examples before answering
  - Chain-of-Thought (CoT): Step-by-step reasoning process
  - Self-Consistency: Generates 3 different answers and measures consistency
  - CoT + Self-Consistency: Combines reasoning steps with multiple samples

2. Multi Agent

- 3 personalities:
  - Internal Medicine specialist
  - Emergency Medicine specialist
  - Family Medicine specialist

3. Conventional Rule-Based System

- Pattern Matching: Uses predefined medical keyword patterns
- Rule Database: Maps symptoms/conditions to standard treatments
- Extremely fast but not efficient.

Method	Strength	Best use	Limits	Typical latency*
Rule-Based	Deterministic, instant	Definitions/patterns	Narrow coverage	~instant
Single-LLM	Fluent, adaptable	General QA	Calibration, hallucination risk	low–moderate
Multi-Agent	Conservative consensus	Ambiguous/high-stakes	Higher cost/latency	higher

LLM Model - llama3.1-8b-local

Input: Medical question(here in this case, what is diabetes? what are its symptoms?)

Output: Relavent response from methods

```
🔷 LLM (ALL TECHNIQUES)
   Answer:
   Time: 245.31s
   Confidence: High (Multiple techniques)
   Reasoning: Compared all prompt techniques and selected cot_self_consistency (most thorough)

🔷 MULTI-AGENT (MEDICAL TEAM)
   Answer: After reviewing the opinions from our Internal Medicine, Emergency Medicine, and Family Medicine specialists, I'd like to provide a comprehensive final re
commendation on diabetes.

**Definition:** Diabetes mellitus is a chronic metabolic disorder characterized by high blood sugar levels (hyperglycemia) due to defects in insulin secretion, insul
in action, or both. Insulin is a hormone produced by the pancreas that regulates glucose uptake into cells throughout the body.

**Types of Diabetes:**

There are primarily two types of diabetes:

1. **Type 1 Diabetes**: An autoimmune disease where the immune system attacks and destroys the insulin-producing beta cells in the pancreas, resulting in a complete
deficiency of insulin production.
2. **Type 2 Diabetes**: The most common form of diabetes, characterized by insulin resistance (when the body's cells become less responsive to insulin) or impaired i
nsulin secretion.

**Symptoms:**

While some people with diabetes may not experience any symptoms at all, common signs and symptoms include:

* Increased thirst and
   Time: 65.79s
   Confidence: Very High (Multiple expert consensus)
   Reasoning: Three medical specialists collaborated and coordinated final recommendation

🔷 CONVENTIONAL (RULE-BASED)
   Answer: Based on the symptoms 'diabetes', common medical considerations include: blood sugar control, metformin, insulin, glucose monitoring. Please consult with
a healthcare provider for proper evaluation and treatment.
   Time: 0.00s
   Confidence: Medium (Pattern matching)
   Reasoning: Matched medical pattern 'diabetes' with known treatments/considerations
```

FUTURE WORK

1. Agentic Workflow Development

- Connect summarizer, diagnosis, Q&A, and risk modules.
- Orchestrate through LangGraph framework.

2. Heterogeneous RAG System

- Integrate multiple datasets into a single retrieval pipeline.
- BigQuery for structured patient data.
- Vertex AI Vector Index for unstructured notes and documents.

3**. Model Deployment**

- Deploy all fine-tuned models on Vertex AI.
- Expose secure model endpoints for downstream use.

4. Observability & Monitoring

- Implement LangSmith for evaluations, traces, and error analysis.
- Use Cloud Logging & Monitoring for system health

THANK YOU!